

JUAN IGNACIO OGDON

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Clickable links: [LinkedIn](#) | [GitHub Portfolio](#)



PROFILE

Aerospace Engineering student (6th year, expected graduation Dec 2026) at UTN, focused on control systems, genetic algorithms, and neural network-based controllers for autonomous aerospace applications. Experienced in spacecraft mechanisms, rocket simulation, and international research collaboration. Seeking graduate research opportunities in advanced control and AI-based systems. I invite you to know more about all my projects on my portfolio or my research papers on ResearchGate (links throughout resume).

EDUCATION

- Universidad Tecnológica Nacional (UTN)** Expected Dec 2026
B.Sc. Aerospace Engineering (6-year degree) | avg grade: 8.1/10 Buenos Aires, Argentina
- Don Bosco High School** Dec 2020
Diploma in Economics and Administration | avg grade: 9.24/10 San Carlos de Bariloche, Argentina

EXPERIENCE

- Universidad Tecnológica Nacional (UTN)** 2025 – Present
Teaching Assistant — Control Systems (click to see more) Buenos Aires, Argentina
– Focused on control systems for aerospace applications, with emphasis on AI-based control methods, simulation, and system dynamics.
- Private Tutor — Mathematics & Physics (click to see more)** 2019 – Present
Independent Buenos Aires, Argentina
– One-on-one tutoring over 7+ years; sustained student referrals reflect consistent teaching effectiveness.
– Supported students with diverse learning needs, helping them achieve significant academic improvements.
- LIA Aerospace** 2024 – 2025
Intern Buenos Aires, Argentina
– Worked designing ground support equipment for liquid rocket engine.
- Grupo de Tecnología Aeroespacial, UTN** 2022 – 2024
Research Student Buenos Aires, Argentina
– Developed a CubeSat appendage release system for solar panels; awarded **Best Presentation in Satellite Technology** at the IAA Latin American Small Satellite Conference (2024).

SELECTED PROJECTS (CLICK TO SEE MORE)

- GA-Evolved Neural Network Rocket Landing Controller** — Trained a minimal neural network to solve 2D TVC rocket landing using a genetic algorithm pipeline. Features adaptive curriculum staging, island-model population diversity, and physics-derived anti-exploit fitness design across three rocket families. [ResearchGate Paper](#)
- CubeSat Appendage Release System** — Design, simulation, and characterization of a solar panel deployment mechanism; awarded Best Presentation at IAA 2024. [ResearchGate Paper](#)
- Rocket Attitude Control Simulation** — Simulated thrust vector control rocket dynamics from scratch using Simulink, with Guidance and control algorithms.

SKILLS

- Programming & Simulation:** MATLAB, Simulink, Mathematica, Python, OpenFOAM
- Engineering Tools:** CAD Design, FDM 3D Printing, Optimized Design for 3D Printing, FEA
- Collaboration:** Atlassian Jira, Confluence; team-based research projects
- Soft Skills:** Strong communicator and collaborative problem-solver; adaptable across research, engineering, and teaching environments; quick, self-directed learner.

LANGUAGES

- Spanish: Native | English: Bilingual (C1 certified)